Reef Rescue

Perception-guided coral handling, with a reviewable evidence trail.

THE SCENARIO

For reef-restoration teams, the prototype inspects a registered, already-detached coral fragment, transports it to an eligible bleached recipient and verifies mechanical attachment. Supplied registry facts establish species, lineage and eligibility. The loose-fragment workflow does not cut a healthy colony.

1 / OBSERVE	2 / DECIDE	3 / ACT & VERIFY
Rendered robot-camera RGB **Gemma appearance evidence**	Fresh evidence + registry **Local policy + native cuOpt**	Browser safety and controller **Collect, place, verify**

Uncertainty: hold and ask Argon for **rescan, hold or abort**. Argon supplies bounded context and visual checks for Gemma's next fresh image. Registry and local handling gates remain authoritative; Argon receives structured observations.

MODELS AND AUTHORITY

Component / exact identifier	Execution and responsibility
Gemma gemma-4-E4B-it	Weights: google/gemma-4-E4B-it. Hosted vLLM on Modal; separately measured on Spark GB10 (page 2). Reports appearance, target visibility and fragment/gripper state.
Gemini Argon gemini-4-argon	Google Interactions API. Bounded held-issue advice and optional terminal review. No automatic code changes or deployment.
NVIDIA cuOpt nvidia-cuopt	Native route evaluation on Modal; recorded solver 26.10.0. Chooses among supplied alternatives, subject to local validation.
Policy / controller reef-local-policy-v1	Inspectable Python gateway policy; browser mission controller, reduced physics and safety checks remain authoritative.

MEASURED HOSTED EPISODE [1]

1 verified placement in **80.55 simulated seconds**; 0 collisions, 0 drops and 0 fragment recoveries. 5 camera responses, uncertain evidence, Argon-assisted rescan and a sensor-dropout hold precede successful placement.

Measurement	Observed value	Scope
Gemma request latency	2.687-3.201 s; n=5	Browser round trip
Argon assistance latency	22.700 s; n=1	Browser round trip
cuOpt solve time	0.198 s; n=1	Native solver duration

Run the pipeline. Configure Gemma, native cuOpt and Argon on the gateway. Load the compact scene through Environment builder, select Robot + base, wait for service readiness and start the mission. Export captures, provider receipts and outcome counters together. [3]

REPRODUCE AND AUDIT

Run `python3 sim/pelagic/scripts/run_lab.py`; configure actual services using SETUP.md. Verify with `python3 tools/verify_submission.py` and the documented test suites.

[1] docs/evidence/live-model-episode.json - 2026-11-08 13:41:14 UTC; SHA-256 18ccaa83ff528e05...
[2] docs/EDGE_DEPLOYMENT.md and docs/evidence/edge_gemma_benchmark.json - SHA-256 4516a81d6166dd8e...
[3] SETUP.md and docs/IMPLEMENTATION_STATUS.md. Artifact format and hashes: docs/EVIDENCE.md; rubric: docs/JUDGING_GUIDE.md.

Measured edge operation

Pinned Gemma inference and native route services on NVIDIA GB10.

5/5 valid structured image responses on NVIDIA GB10. The Spark container used Docker network mode **none**, with only loopback and no network routes. Model weights were downloaded beforehand. [2]

OBSERVED RESOURCE BUDGET [2]

Measurement	Result and interpretation
Image request latency	Median 5.150 s; range 5.052-7.793 s. 5 repeats of one saved Isaac-rendered image; first-use compilation and warm caches affect timing.
Container memory	Lifetime cgroup peak 18.397 GiB; peak sampled memory.current 6.449 GiB. These are not total unified CPU + GPU memory.
Configured limits	48 GiB memory, 8 CPU cores, BF16, one request sequence, 4,096-token context, eager execution. Limits are not measurements.
Shared GPU power	14.94-31.92 W; sample mean 28.735 W across 127 samples. Isaac and other processes shared the GPU; this is not isolated Gemma power or vehicle energy.
Pinned model / runtime	google/gemma-4-E4B-it; revision 4ecc2ba63bca... vLLM 0.21.0+2325b6f0.nvinternal.26.5.post1.53039255; Transformers 5.6.0. Full image digest and metadata: [2].

The five image requests used the same recorded input and returned an **unknown gripper state**. The controller requires a matching visible fragment and gripper state before pickup. The browser gateway connects local Gemma and cuOpt with online Argon assistance.

JUDGING CRITERIA: EVIDENCE A REVIEWER CAN INSPECT

Criterion	Implementation and evidence
25% / Perception-driven action	Image hashes, capture/target IDs and freshness gates connect Gemma observations to action. Appearance cannot establish viability or genotype.
25% / Adaptation to change	Actual uncertain-view rescan and sensor-dropout hold in [1]. Current-induced loss and same-ID recovery are covered by controller/model-double regression tests.
15% / Operational autonomy	Local safety continues during remote waits; assistance retains a 90-second overall hold and three-attempt limit. Disconnected Gemma image inference is measured in [2].
20% / Relevance and impact	Registered fragment handling targets a concrete restoration workflow. Identity, damage and attachment checks are inspectable; ecological benefit and survival are not established.
15% / Technical quality	Native cuOpt, explicit model provenance, cross-language contract tests, bounded memory and telemetry. Independent Isaac RTX feed is read-only; browser physics is separate.

Review and operation: docs/CODE_MAP.md explains the files and entry points. SETUP.md provides service configuration and test commands. docs/EVIDENCE.md explains episode fields and source hashes; docs/IMPLEMENTATION_STATUS.md describes mission execution and service behavior.